

Osterquill Toxicology Inc

AP02 reserve analogues: integrated hERG and CYP screen, 44 compounds

Final Report

Study OT-23-S0046 (LIMS OQ/2023/000755)

Client: Avernoelwick Pharmaceuticals Ltd

Study lead: Dr N. Takamori, Head of Toxicology

Report date: March 23, 2023

1 Summary

44 compounds from the Avernoelwick aminopyrimidine regioisomer series 02 were tested for Avernoelwick Pharmaceuticals Ltd in the hERG (IKr) safety pharmacology, CYP safety interaction panel. Testing started on March 8, 2023 and results were reported on March 23, 2023. All accepted runs met control validity criteria; one run was rejected and repeated (see Section 6). No compound had a hERG IC₅₀ below 1 µM. Most compounds (25 of 44) had IC₅₀ at or above 10 µM or above the top concentration. Inhibition above 50% at 10 µM was most frequent for CYP3A4; CYP1A2 was rarely affected.

Table 1 Study summary

Metric	Value
Project	OT-23-S0046
Client	Avernoelwick Pharmaceuticals Ltd
Compounds tested	44
Series / chemotype	Avernoelwick aminopyrimidine regioisomer series 02 (Aminopyrimidine regioisomer)
Assays	hERG patch clamp, CYP inhibition panel (1A2/2C9/2C19/2D6/3A4) at 10 µM
Turnaround (calendar days)	15
Price	USD 25,000
hERG IC ₅₀ <1 µM	0
hERG IC ₅₀ 1-10 µM	19
hERG IC ₅₀ ≥10 µM	25
hERG median IC ₅₀ (µM)	12
E-4031 control IC ₅₀ (nM)	15
CYP1A2 >50% inhibition @10 µM	2
CYP2C9 >50% inhibition @10 µM	3
CYP2C19 >50% inhibition @10 µM	11
CYP2D6 >50% inhibition @10 µM	8
CYP3A4 >50% inhibition @10 µM	13
Compounds with any CYP >50%	19
Ketoconazole (3A4) IC ₅₀ (µM)	0.023

Compliance statement

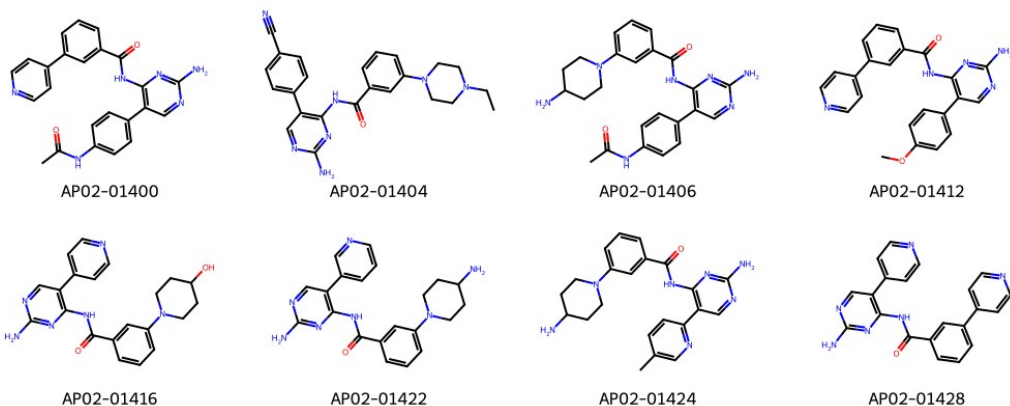
This exploratory in vitro investigation is not conducted under GLP. Procedures follow Osterquill standard operating procedures, and Quality Assurance audits the data for integrity only; no GLP compliance statement is made.

2 Objective

To rank the test article cohort for hERG (IKr) safety pharmacology, CYP safety interaction panel liability so that Avelon Pharmaceuticals Ltd could choose analogues for progression.

3 Test items

44 compounds were received as 10 mM DMSO solutions in one test article cohort on March 8, 2023 and logged against OQ/2023/000755. Identity and concentration were taken as supplied by the client. Stocks were stored at -20 °C and thawed once for testing.



4 Methods

hERG (IKr) safety pharmacology

HEK293 cells stably expressing hERG (KCNH2) were recorded in whole-cell configuration on a 16-channel automated patch clamp platform at room temperature. Cells are held at -80 mV, depolarized to +40 mV for 2 s and repolarized to -50 mV for 2 s to elicit the tail current, at 0.1 Hz. Each compound was applied cumulatively at 0.12, 0.37, 1.1, 3.3, 10 and 30 μ M (0.3% DMSO final). Each concentration is held until the tail-current block reaches a steady state (change below 5% over the last three sweeps), with a minimum of 3 and a maximum of 8 minutes; compounds still drifting at 8 minutes are flagged as slow-onset.

E-4031 is run as a separate 6-point concentration response (1-300 nM) on every run.

- Control validity: seal resistance ≥ 500 M Ω before break-in and ≥ 200 M Ω throughout
- Tail current ≥ 200 pA at baseline; run-down $\leq 10\%$ over the vehicle period
- E-4031 IC₅₀ within 5-50 nM; runs outside this range are rejected and repeated
- IC₅₀ fitted per cell (four-parameter logistic); the compound IC₅₀ is the geometric mean of finite cell IC₅₀s when these are the majority of accepted cells, otherwise reported as >30 μ M
- Where precipitation is seen at 30 μ M, that concentration is excluded and the highest evaluable concentration is reported

CYP safety interaction panel

Pooled human liver microsomes (0.1 mg/mL protein) were incubated with each compound at 10 μ M and an isoform-selective probe substrate at approximately K_m, in duplicate, for 10 minutes at 37 °C after NADPH addition. Metabolite formation is measured by LC-MS/MS against vehicle (0.1% DMSO) incubations; the lower limit of quantitation of each metabolite is at least 20-fold below the vehicle signal.

Probe substrates and controls

Isoform	Probe substrate	Metabolite	Positive control
CYP1A2	Phenacetin	Acetaminophen	Furafylline
CYP2C9	Diclofenac	4'-Hydroxydiclofenac	Sulfaphenazole
CYP2C19	S-Mephenytoin	4'-Hydroxymephenytoin	Ticlopidine
CYP2D6	Dextromethorphan	Dextrorphan	Quinidine
CYP3A4	Midazolam	1'-Hydroxymidazolam	Ketoconazole

Positive controls are run as 7-point concentration responses on every run. Furafylline and ticlopidine (mechanism-based inhibitors) are preincubated with microsomes and NADPH for 15 minutes before probe addition. A run is accepted when each control IC₅₀ falls within 0.5-2 times the laboratory historical mean. Results are reported as mean percent inhibition relative to vehicle; values above 50% at 10 μ M are flagged, and small negative values indicate no inhibition.

5 Results

hERG results

All 44 compounds were tested. No compound had a hERG IC₅₀ below 1 μ M. Most compounds (25 of 44) had IC₅₀ at or above 10 μ M or above the top concentration. The median IC₅₀ was 12 μ M (0 below 1 μ M, 19 between 1 and 10 μ M, 25 at or above 10 μ M, of which 4 above the highest evaluable concentration). E-4031 gave an IC₅₀ of 15 nM in the accepted runs.

Compounds with QC notes: AP02-01566: First run rejected (E-4031 outside range); repeated; AP02-01590: Precipitation at 30 μ M; 30 μ M point excluded from fits. Values for these compounds should be read with the note.

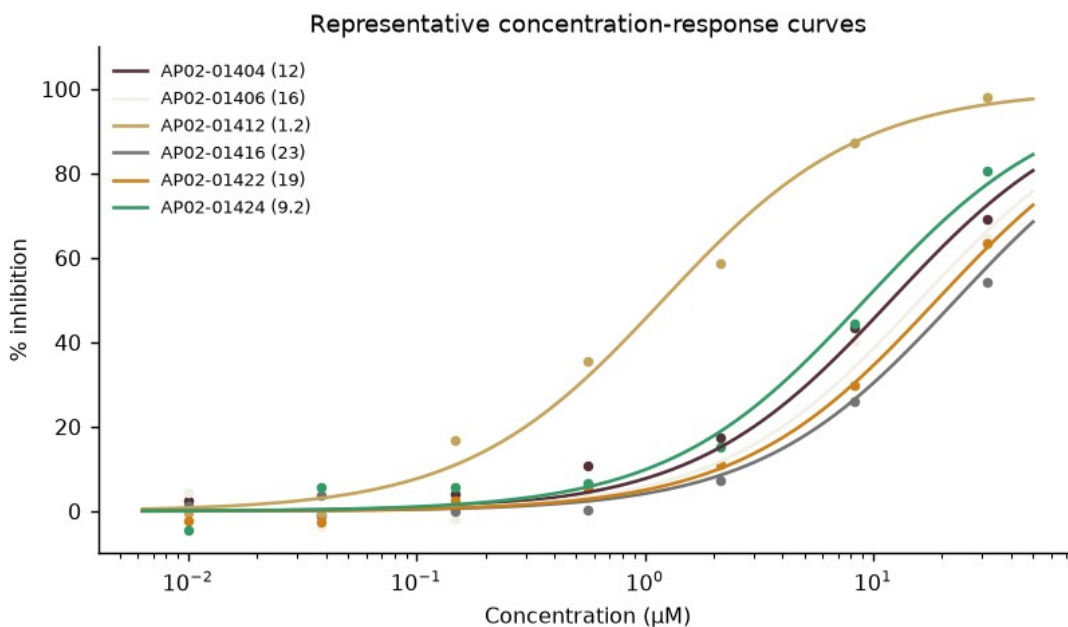
Table 2 hERG IC₅₀ by compound (μ M)

Compound	IC ₅₀ cell 1 (μ M)	IC ₅₀ cell 2 (μ M)	IC ₅₀ cell 3 (μ M)	IC ₅₀ geomean (μ M)	% inhibition @10 μ M	n cells (finite)	QC note
AP02-01400	>30	>30		>30	26	2 (0)	
AP02-01404	12	6.8	8.5	8.9	52	3 (3)	
AP02-01406	16	9.9		13	45	2 (2)	

AP02-01412	1.2	1.6		1.4	85	2 (2)	
AP02-01416	23	24		24	25	2 (2)	
AP02-01422	19	17		18	38	2 (2)	
AP02-01424	9.2	11	9.7	10	43	3 (3)	
AP02-01428	28	>30		>30	17	2 (1)	
AP02-01436	22	16		19	30	2 (2)	
AP02-01440	14	11		12	47	2 (2)	
AP02-01446	15	14		15	42	2 (2)	
AP02-01448	4.1	3.5		3.8	75	2 (2)	
AP02-01452	16	22		18	39	2 (2)	
AP02-01458	5.0	4.9		4.9	70	2 (2)	
AP02-01464	15	14		15	44	2 (2)	
AP02-01470	6.1	4.8		5.4	67	2 (2)	
AP02-01472	21	13		16	37	2 (2)	
AP02-01476	5.7	9.6		7.4	56	2 (2)	
AP02-01478	2.5	3.5		3.0	80	2 (2)	
AP02-01484	7.7	15		11	47	2 (2)	
AP02-01490	>30	26	>30	>30	22	3 (1)	
AP02-01494	3.6	2.7		3.1	77	2 (2)	
AP02-01496	30	24		27	30	2 (2)	
AP02-01500	6.4	4.9		5.6	64	2 (2)	
AP02-01502	8.7	9.4		9.1	52	2 (2)	
AP02-01508	16	12		14	44	2 (2)	
AP02-01514	12	21		16	34	2 (2)	
AP02-01518	2.3	2.4		2.4	82	2 (2)	
AP02-01520	24	>30		>30	26	2 (1)	
AP02-01530	16	14		15	37	2 (2)	
AP02-01532	5.4	4.5		5.0	68	2 (2)	
AP02-01536	4.4	5.0		4.7	63	2 (2)	
AP02-01538	21	11		15	44	2 (2)	
AP02-01542	10	6.6	11	9.2	49	3 (3)	
AP02-01550	8.2	10		9.2	53	2 (2)	
AP02-01554	25	28	>30	27	29	3 (2)	
AP02-01556	6.8	7.2		7.0	63	2 (2)	
AP02-01560	8.1	6.2		7.1	60	2 (2)	
AP02-01566	12	15		13	45	2 (2)	First run rejected (E-4031 outside range); repeated
AP02-01568	12	15		13	42	2 (2)	
AP02-01572	6.2	4.6		5.3	66	2 (2)	
AP02-01574	22	16		19	38	2 (2)	
AP02-01584	9.0	11		10	41	2 (2)	
AP02-01590	10	8.1		9.0	56	2 (2)	Precipitation at 30 μ M; 30 μ M point excluded from fits

Top concentration 30 μ M; 6-point, 3-fold dilution; positive control E-4031 IC50 15 nM (acceptance 5-50 nM)

Compound IC50 = geometric mean of finite cell IC50s when they are the majority of accepted cells; otherwise reported as >30 μ M (or >10 μ M where solubility-limited)



CYP inhibition results

Inhibition above 50% at 10 µM was most frequent for CYP3A4; CYP1A2 was rarely affected. In total 19 compounds inhibited at least one isoform by more than 50% at 10 µM (CYP3A4: 13; CYP2D6: 8; CYP2C19: 11; CYP2C9: 3; CYP1A2: 2). All control IC₅₀s were within 0.5-2 times the historical mean (ketoconazole 0.023 µM against a historical mean of 0.030 µM).

Table 3 Percent inhibition at 10 µM

Compound	CYP1A2 (%)	CYP2C9 (%)	CYP2C19 (%)	CYP2D6 (%)	CYP3A4 (%)
AP02-01400	13	10	51	22	58
AP02-01404	1	5	9	5	16
AP02-01406	1	-3	9	-1	-3
AP02-01412	38	86	94	45	98
AP02-01416	1	-2	-6	1	7
AP02-01422	-2	0	9	3	-0
AP02-01424	4	-4	4	4	-2
AP02-01428	54	5	43	8	36
AP02-01436	8	1	7	12	-2
AP02-01440	-8	-3	3	14	-1
AP02-01446	15	84	90	6	93
AP02-01448	-6	-1	1	6	68
AP02-01452	3	22	73	10	12
AP02-01458	14	43	77	47	23
AP02-01464	5	1	-3	6	8
AP02-01470	2	5	1	5	-2
AP02-01472	-6	2	5	19	41
AP02-01476	5	19	50	66	39
AP02-01478	27	17	65	76	51
AP02-01484	-1	2	29	1	8
AP02-01490	-3	8	-1	4	2
AP02-01494	46	22	68	14	61

AP02-01496	-11	-0	-1	4	8
AP02-01500	-3	1	1	3	8
AP02-01502	3	-7	16	1	6
AP02-01508	2	43	13	95	72
AP02-01514	-2	9	48	4	7
AP02-01518	6	18	87	89	65
AP02-01520	7	12	20	9	45
AP02-01530	3	0	1	6	16
AP02-01532	2	12	9	68	33
AP02-01536	4	5	80	77	72
AP02-01538	0	1	11	51	-4
AP02-01542	10	8	4	2	-1
AP02-01550	2	-7	17	4	8
AP02-01554	71	48	81	86	89
AP02-01556	-0	69	91	47	73
AP02-01560	-2	3	-2	-5	6
AP02-01566	3	-6	29	3	76
AP02-01568	-3	8	17	35	34
AP02-01572	1	12	18	20	14
AP02-01574	28	0	35	4	87
AP02-01584	-1	2	-6	8	24
AP02-01590	19	13	40	5	47

Single concentration 10 μ M, duplicate incubations, mean reported; negative values = no inhibition

Controls (IC₅₀, μ M): furafylline (CYP1A2) 1.7; sulfaphenazole (CYP2C9) 0.33; ticlopidine (CYP2C19) 1.2; quinidine (CYP2D6) 0.042; ketoconazole (CYP3A4) 0.023

Table 4 Positive control IC₅₀s

Control	Isoform	IC ₅₀ (μ M)	Historical mean (μ M)
Furafylline	CYP1A2	1.7	1.8
Sulfaphenazole	CYP2C9	0.33	0.35
Ticlopidine	CYP2C19	1.2	1.2
Quinidine	CYP2D6	0.042	0.060
Ketoconazole	CYP3A4	0.023	0.030

6 Deviations

The first hERG run for one compound was rejected because E-4031 fell outside its acceptance range; the compound was retested in a later run and only accepted data are reported. Precipitation at the top concentration is noted against the affected compounds in Table 2.

7 Discussion

No compound had a hERG IC₅₀ below 1 μ M. Most compounds (25 of 44) had IC₅₀ at or above 10 μ M or above the top concentration. Inhibition above 50% at 10 μ M was most frequent for CYP3A4; CYP1A2 was rarely affected. Results are exploratory and support compound selection; they are not a substitute for a definitive GLP study.

8 Conclusion

Recommended next steps: prioritize compounds with hERG IC50 at or above 10 μ M; for any compound below 10 μ M, compare the IC50 with projected free plasma exposure before progression; repeat the flagged hERG measurements if a selection decision depends on them; run concentration responses for compounds flagged on CYP3A4 and any other isoform above 50%.

9 Records

Raw data, observation sheets and instrument files are retained for 10 years under OQ/2023/000755.

Name	Role	Signature	Date
Naomi Takamori	Head of Toxicology	<u>N Takamori</u>	March 23, 2023
Lucia Moretti	GLP Quality Assurance and Archive Lead	<u>L Moretti</u>	March 22, 2023